

IE02

Measurement of Development: HDI, Indices, Poverty and Inequality

2-3 questions a paper, and the highest repeat density in the entire Economy section. The World Happiness Report came in 2021, 2023 AND 2024 - three consecutive papers. HDI components came in 2024. Ease of Doing Business in 2021 and 2024. Work Participation Ratio in 2021 and 2024. This is a ranking-and-publisher chapter: learn who publishes what, and India's latest rank with its edition year.

Once you accept that development is more than output, you need numbers for it. This chapter is the toolbox - HDI, MPI, PQLI, GII, the global rankings, the poverty lines and the labour force measures. It is the most factual chapter in the Economy paper and the most reliably repeated.

Measuring Development

HDI family (UNDP)

- HDI - 3 dimensions, 4 indicators, geometric mean since 2010
- IHDI - inequality-adjusted (2010)
- GDI - female HDI / male HDI
- GII - lower value is better
- HDR 2025: India 130/193, HDI 0.685, medium; Iceland 1st

Poverty measures

- Global MPI - UNDP + OPHI, 3 dimensions, 10 indicators
- National MPI - NITI Aayog, 3 dimensions, 12 indicators
- Alkire-Foster: $MPI = H \times A$
- Committees: Lakdawala 1993, Tendulkar 2009, Rangarajan 2014

Global rankings

- World Happiness Report - Gallup, Cantril ladder
- Global Hunger Index - Concern Worldwide + Welthungerhilfe
- Global Innovation Index - WIPO
- Doing Business - World Bank, discontinued Sept 2021
- BRAP state rankings - DPIIT

Inequality

- Lorenz curve - Max O. Lorenz, 1905
- Gini coefficient - 0 equality to 1 inequality
- Kuznets curve - inverted U, Simon Kuznets 1955
- Environmental Kuznets Curve

Employment

- PLFS - NSO, launched April 2017
- LFPR, WPR, Unemployment Rate
- Usual Status (ps+ss) and Current Weekly Status
- PLFS 2025: LFPR 59.3, WPR 57.4, UR 3.1

Why composite indices exist

Per capita income tells you the average size of the wallet. It does not tell you whether a child survives infancy, finishes school, or eats twice a day. So from the 1970s economists began building composite indices - one number stitched from several social indicators. The exam does not ask you to evaluate them. It asks four things about each index: who publishes it, what its components are, what its scale means, and India's latest rank. Organise your revision around those four questions and nothing in this chapter can surprise you.

- A composite index combines several indicators into one score, usually normalised to 0-1 or 0-100.
- Income-based measures show the average; social indices show the distribution of wellbeing.
- Rankings move every year - always quote the EDITION year with the rank.
- Where a lower value is better (GII, Gini), say so explicitly - RPSC exploits this.

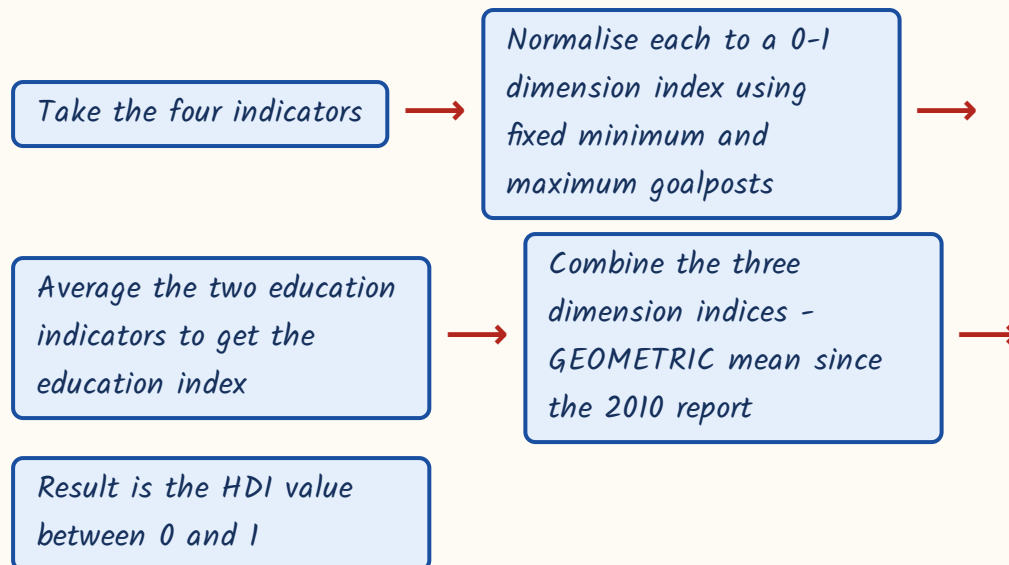
The Human Development Index - the spine of the chapter

The HDI was conceived by Mahbub ul Haq with Amartya Sen's assistance, and UNDP published the first Human Development Report in 1990. Its logic is simple: a good life needs a long life, knowledge, and enough money. So three dimensions - health, education, standard of living. But note the count carefully. Three DIMENSIONS, four INDICATORS, because education is measured by two separate indicators.

HDI - 3 dimensions, 4 indicators

Dimension	Indicator(s)
Health / long life	Life expectancy at birth
Knowledge / education	Mean years of schooling (adults 25+) AND expected years of schooling (children)
Standard of living	Gross National Income (GNI) per capita at purchasing power parity

How the HDI is computed



- Before 2010 the HDI used an ARITHMETIC mean; since 2010 it is the GEOMETRIC mean.
- The geometric mean punishes imbalance - a country cannot buy its way past poor health with high income.
- HDI classes: Very high 0.800 and above; High 0.700-0.799; Medium 0.550-0.699; Low below 0.550.
- Human Development Report 2025 (released May 2025), theme - 'A matter of choice: people and possibilities in the age of AI'.
- India: rank 130 out of 193 countries, HDI value 0.685 - the MEDIUM human development category.
- Iceland ranked first in HDR 2025.

ASKED IN RAS

Asked in RAS Pre 2024 - which indicator is used for standard of living in the HDI? Answer: Gross National Income per capita at purchasing power parity. Not GDP per capita, not per capita consumption.

ASKED IN RAS

Asked in RAS Pre 2024 in negative form - which of the following is NOT a component of the Human Development Index? Answer: infant mortality rate. IMR belongs to PQLI, not HDI. This is the most common single confusion in the chapter.

Two counting points before you leave this section. Three dimensions but FOUR indicators - education carries two. And India's HDI value of 0.685 sits in the MEDIUM band even though a rank of 130 out of 193 sounds worse than that; rank and category are separate questions, so answer whichever one the stem asks.

The rest of the UNDP family, plus PQLI and GNH

Around the HDI, UNDP built a set of adjusted versions. Each one takes the same idea and asks a further question - what if we account for inequality, what about the gap between men and women, what about deprivation rather than averages. Learn them as one-line definitions; that is exactly how they are matched in the exam.

The UNDP index family - one line each

Index	What it does
HDI	Average achievement in health, education and income
IHDI	Inequality-adjusted HDI, introduced 2010; equals HDI only when there is no inequality
GDI	Gender Development Index - the ratio of female HDI to male HDI
GII	Gender Inequality Index - reproductive health, empowerment, labour market; LOWER value is better
MPI	Multidimensional Poverty Index - counts overlapping deprivations, not income

- PQLI - Physical Quality of Life Index, given by Morris David Morris in 1979.
- PQLI has exactly three indicators: infant mortality rate, life expectancy at AGE ONE, and basic literacy rate.
- PQLI is scaled 0 to 100 and deliberately EXCLUDES income - that exclusion is its signature.
- Gross National Happiness is Bhutan's development philosophy - four pillars and nine domains.
- Chronology of first introduction: PQLI 1979, HDI 1990, then IHDI/GII 2010, then the Global MPI.

TRAP

Life expectancy at BIRTH is HDI. Life expectancy at AGE ONE is PQLI. One word separates the two indices, and RPSC has built a question on exactly that word.

Multidimensional poverty - global MPI and India's national MPI

Income poverty asks one question - is your money below a line.

Multidimensional poverty asks whether you are simultaneously deprived in several basic things at once: no schooling, no toilet, no clean fuel, no assets.

The method is the Alkire-Foster method, and the formula is short enough to memorise: $MPI = H \times A$, where H is the headcount ratio (how many people are poor) and A is the intensity (how many deprivations the average poor person suffers).

Global MPI vs NITI Aayog's National MPI

Global MPI	National MPI (India)
Published jointly by UNDP and OPHI (Oxford Poverty and Human Development Initiative)	Published by NITI Aayog
3 dimensions, 10 indicators	3 dimensions, 12 indicators
Health, education, standard of living	Same three dimensions
-	Adds two indicators: maternal health and bank account
Alkire-Foster method, $MPI = H \times A$	Same Alkire-Foster method

- A person is multidimensionally poor if deprived in at least ONE-THIRD (33.33%) of the weighted indicators.
- NITI Aayog's 'National MPI: A Progress Review 2023' - multidimensional poverty fell from 24.85% (2015-16) to 14.96% (2019-21).
- MPI is NOT published by the IMF or the World Bank - this is a standard wrong-pair option.
- Per capita income is NOT an indicator in the National MPI; the twelve indicators are all deprivation-based.

The global index scoreboard - publisher and India's rank

This is the table to photograph in your mind. Every year RPSC picks two or three of these and asks either the publishing agency or India's rank. Note the editions - a rank without an edition year is worthless, because they change annually.

Index to publisher to India's latest position

Index	Published by	India (latest, dated)
Human Development Index	UNDP	130 of 193, HDI 0.685 - HDR 2025 (May 2025)

Index	Published by	India (latest, dated)
World Happiness Report	Gallup / WHR editorial board (UN SDSN)	116 of 147 - 2026 edition (March 2026)
Global Hunger Index	Concern Worldwide (Ireland) + Welthungerhilfe (Germany)	102nd - GHI 2025
Global Innovation Index	World Intellectual Property Organization (WIPO)	38th - GII 2025
Global MPI	UNDP + OPHI	3 dimensions, 10 indicators
Doing Business (Ease of Doing Business)	World Bank - DISCONTINUED September 2021	63 of 190 - Doing Business 2020 (last ever rank)

- The Doing Business report was discontinued in September 2021 after a data-irregularity review.
- Its replacement is the World Bank's Business Ready (B-READY) report.
- Inside India, DPIIT runs the Business Reform Action Plan (BRAP) and ranks States and UTs on ease of doing business.
- DPIIT = Department for Promotion of Industry and Internal Trade, Ministry of Commerce and Industry.
- GHI's four indicators: undernourishment, child stunting, child wasting, child mortality.

ASKED IN RAS

Asked in RAS Pre 2021 and again in 2024 - the Ease of Doing Business ranking of countries was published by the World Bank in a report discontinued in 2021. Its title? Answer: Doing Business. Learn the pair 'World Bank - Doing Business - discontinued Sept 2021 - India 63rd in DB 2020'.

YAAD RAKHO

Yaad rakho the publishers as a chain: Hunger is a CHARITY pair (Concern Worldwide + Welthungerhilfe), Innovation is a PATENT body (WIPO), Human Development and MPI are UNDP, Doing Business was the WORLD BANK. Once the publisher is fixed, half the questions in this chapter answer themselves.

The World Happiness Report - three consecutive papers, so know it cold

Give this one its own revision slot. RPSC asked it in 2021, in 2023 and again in 2024 - three papers in a row. Nothing else in the Economy paper repeats like that. The report ranks countries on how people rate their own lives, not on any objective statistic. That is the conceptual point examiners test.

- Data source: the Gallup World Poll.
- Method: self-reported life evaluations on the CANTRIL LADDER, a 0 to 10 scale.
- Six explanatory variables: GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, perceptions of corruption.
- These six EXPLAIN the score; they do not construct it. The score itself is the survey answer.
- World Happiness Report 2026 (March 2026): Finland first for the NINTH consecutive year; Afghanistan last.
- India: 116 out of 147 countries in the 2026 edition.

India's World Happiness Report rank by edition

<i>Edition</i>	<i>India's rank</i>
2021	139
2023	126
2024	126
2025	118
2026	116 of 147

ASKED IN RAS

Asked in RAS Pre 2021, 2023 AND 2024 - India's rank in the World Happiness Report. For the 2026 edition the answer is 116th. Whichever edition the paper names, read the year in the question stem before choosing.

ASKED IN RAS

Asked in RAS Pre 2023 - which country has been ranked happiest for consecutive years? Answer: Finland, and in the 2026 edition it is the ninth consecutive year. Also asked as a statement set in 2021 and 2024: the report is based on Gallup World Poll data - correct.

TRAP

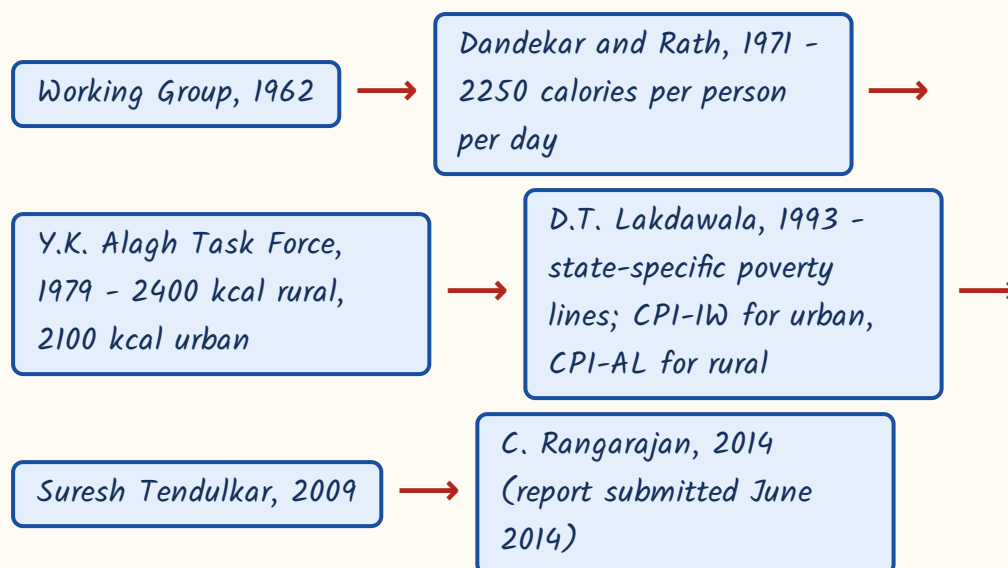
Happiness rank is out of 147 countries; HDI rank is out of 193. Different denominators, and options are written using the wrong one. Also, the WHR is not a UNDP product - UNDP does HDI and MPI.

Poverty in India - the committee chain

India's poverty line has been redrawn roughly once a decade, and each redrawing carries a name. Learn them as a chronological chain, because RPSC's favourite

format here is 'arrange in chronological order' or a committee-to-year match. Note also what each one changed - calories first, then state-specific lines, then a full basket of goods.

Poverty estimation committees, in order



Tendulkar vs Rangarajan - the two you must be able to quote

Item	Tendulkar (2009)	Rangarajan (2014)
Poverty ratio, 2011-12	21.9%	29.5%
Rural poverty line, per capita per month	Rs 816	Rs 972
Urban poverty line, per capita per month	Rs 1000	Rs 1407
Rural / urban poverty ratio	25.7% rural, 13.7% urban	-

- Alagh 1979 fixed the calorie norms - 2400 rural, 2100 urban. Note the rural norm is HIGHER (harder physical work).
- Lakdawala 1993 is the one that introduced STATE-SPECIFIC poverty lines.
- Rangarajan's higher ratio comes from a broader consumption basket, not from a worsening economy.
- Both Tendulkar and Rangarajan estimates refer to the same year, 2011-12 - that is why they get compared.

YAAD RAKHO

Order mnemonic for the committees: 'Working Dogs Are Loyal, Try Running'
- Working Group, Dandekar-Rath, Alagh, Lakdawala, Tendulkar, Rangarajan.
1962-1971-1979-1993-2009-2014.

Inequality - Lorenz curve, Gini coefficient, Kuznets curve

Inequality is measured by drawing a picture and then measuring the gap inside it. The Lorenz curve plots the cumulative share of income against the cumulative share of population. If income were perfectly equal, the curve would be a 45-degree straight line. The further the actual curve sags away from that line, the more unequal the distribution. The Gini coefficient is simply that sag expressed as a number.

- Lorenz curve - given by Max O. Lorenz in 1905.
- The 45-degree line is the line of perfect equality.
- The further the Lorenz curve lies from the line of equality, the greater the inequality.
- Gini coefficient ranges from 0 (perfect equality) to 1 (perfect inequality).
- Gini is derived from the Lorenz curve - it is the ratio of the gap area to the total area under the equality line.
- Kuznets curve - inverted-U between per capita income and inequality; proposed by Simon Kuznets in 1955.
- Environmental Kuznets Curve applies the same inverted-U shape to income and environmental degradation.

TRAP

Gini measures inequality of DISTRIBUTION, not the level of poverty. A poor country can have a low Gini and a rich one a high Gini. And Gini 0 means perfect equality - a higher Gini is worse, the opposite direction from the HDI.

Unemployment and the PLFS measures

Two things get asked here - the TYPES of unemployment as definitions to match, and the three PLFS ratios as formulas. Take the types first. The one to over-learn is disguised unemployment, because it is the Indian case: more people are working on the farm than the farm needs, so the marginal productivity of the extra worker is near zero. Remove him and output does not fall.

Types of unemployment

Type	Meaning
Structural	Skills or location of workers do not match the jobs the economy now creates - long-term
Frictional	Short transition between jobs; searching, switching
Cyclical	Caused by a downturn in the business cycle; falling aggregate demand
Seasonal	Work available only in part of the year - agriculture, tourism, sugar mills
Disguised	More workers than needed; marginal productivity near zero - typical of Indian agriculture
Educated / open	Qualified people visibly without work

Now the measurement. The Periodic Labour Force Survey replaced the old five-yearly Employment-Unemployment Surveys. It was launched by the NSO in April 2017 and it reports on two reference periods - Usual Status (principal plus subsidiary, a one-year recall) and Current Weekly Status (a seven-day recall).

Three ratios come out of it, and the difference between them is entirely about the DENOMINATOR.

The three PLFS ratios - watch the denominator

Measure	Formula	PLFS 2025 (usual status, 15+)
LFPR	$\text{Labour force} / \text{population} \times 100$	59.3%
WPR (Worker Population Ratio / Work Participation Rate)	$\text{Employed persons} / \text{POPULATION} \times 100$	57.4%
Unemployment Rate	$\text{Unemployed} / \text{LABOUR FORCE} \times 100$	3.1%

- Labour force = employed + unemployed persons actively seeking work.
- WPR is out of the POPULATION; UR is out of the LABOUR FORCE. That is the whole trick.
- PLFS Annual Report 2025 covers January-December 2025 and was released in March 2026.
- Youth (15-29) unemployment rate: 9.9%.
- Male WPR 76.6%, female WPR 38.8% - the gender gap is itself examinable.
- Self-employed 56.2% of workers; regular wage/salaried 23.6%.
- Agriculture's share of employment fell from 44.8% in 2024 to 43.0% in 2025.

ASKED IN RAS

Asked in RAS Pre 2021 and again in 2024 - the Worker Population Ratio is defined as the percentage of EMPLOYED persons in the TOTAL POPULATION. The distractor is 'percentage of the labour force', which is the Unemployment Rate's denominator.

CURRENT LINK

Current link (as of September 2026): the PLFS Annual Report for January-December 2025, released March 2026, is the current employment reference - LFPR 59.3%, WPR 57.4%, unemployment 3.1%. Alongside it, the Human Development Report 2025 (India 130/193), the World Happiness Report 2026 (India 116/147), GHI 2025 (India 102nd) and GII 2025 (India 38th) are the four rankings most likely to appear as current-affairs questions. Carry the edition year with every number.

REVISION RECAP

1. HDI - conceived by Mahbub ul Haq with Amartya Sen; first Human Development Report, UNDP, 1990.
2. HDI = 3 dimensions, 4 indicators; geometric mean since 2010; value 0 to 1.
3. HDI indicators - life expectancy at birth; mean and expected years of schooling; GNI per capita (PPP).
4. HDI bands - Very high 0.800+, High 0.700-0.799, Medium 0.550-0.699, Low below 0.550.
5. HDR 2025 - India 130/193, HDI 0.685, medium category; Iceland first.
6. IHDI inequality-adjusted; GDI = female HDI / male HDI; GII lower is better; MPI counts deprivations.
7. Global MPI - UNDP + OPHI, 3 dimensions 10 indicators; poor if deprived in one-third of weighted indicators.
8. National MPI - NITI Aayog, 3 dimensions 12 indicators (adds maternal health and bank account); $MPI = H \times A$.
9. National MPI Progress Review 2023 - poverty down from 24.85% (2015-16) to 14.96% (2019-21).
10. PQLI - Morris D. Morris 1979; IMR, life expectancy at age one, basic literacy; 0-100; income excluded.
11. World Happiness Report - Gallup World Poll, Cantril ladder 0-10, six explanatory variables.
12. WHR India ranks - 139 (2021), 126 (2023 and 2024), 118 (2025), 116 of 147 (2026). Finland first nine years running.
13. GHI - Concern Worldwide + Welthungerhilfe, four indicators, India 102nd in 2025. GII - WIPO, India 38th in 2025.

14. Doing Business discontinued September 2021; India's last rank 63 of 190 in DB 2020; B-READY replaces it; DPIIT runs BRAP for States.
15. Committee chain - Working Group 1962, Dandekar-Rath 1971, Alagh 1979, Lakdawala 1993, Tendulkar 2009, Rangarajan 2014.
16. Tendulkar - 21.9% in 2011-12, Rs 816 rural / Rs 1000 urban. Rangarajan - 29.5%, Rs 972 rural / Rs 1407 urban.
17. Lorenz curve (Max O. Lorenz, 1905); Gini 0 to 1, derived from it; Kuznets inverted-U, 1955.
18. Disguised unemployment - marginal productivity near zero; the classic Indian agricultural case.
19. PLFS launched by NSO April 2017; Usual Status (ps+ss) and Current Weekly Status.
20. LFPR 59.3, WPR 57.4, UR 3.1 (PLFS 2025). WPR denominator = population; UR denominator = labour force.